

Chapter 10

Circles

Unit 3 — Quadrilaterals and Circles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Circle	Set of all points equidistant from a center
Radius	Segment from center to a point on circle
Chord	Segment with both endpoints on the circle
Diameter	Chord through the center
Secant	Line intersecting a circle at two points
Tangent	Line touching a circle at exactly one point
Arc	Part of a circle
Central angle	Angle whose vertex is the center
Inscribed angle	Angle whose vertex is on the circle

Arc Measures

Arc Type	Symbol	Measure
Minor arc	$\widehat{AB}$	= central angle
Major arc	$\widehat{AXB}$	= $360^\circ - \text{minor arc}$
Semicircle	—	= 180°

Arc Addition: $m\widehat{ABC} = m\widehat{AB} + m\widehat{BC}$

Theorem 10-1: Tangent-Radius Perpendicularity

A **tangent** to a circle is **perpendicular** to the radius drawn to the point of tangency.

$$\overline{OP} \perp \text{tangent at } P$$

Theorem 10-2: Two Tangents from External Point

Two **tangent segments** from the same external point are **congruent**.

$$PA = PB \quad (P \text{ external, } A \text{ and } B \text{ tangency points})$$

Theorem 10-3: Inscribed Angle Theorem

An **inscribed angle** measures **half** its intercepted arc.

$$m\angle ABC = \frac{1}{2}m\widehat{AC}$$

The vertex is on the circle; the sides are chords.

Corollary 10-1: Inscribed Angles — Same Arc

Two **inscribed angles** that intercept the **same arc** are **congruent**.

$$m\widehat{AC} \text{ same} \Rightarrow \angle ABC \cong \angle ADC$$

Corollary 10-2: Inscribed Angle in Semicircle

An angle **inscribed in a semicircle** is a **right angle**.

If $\overline{AC}$ is a diameter $\Rightarrow \angle ABC = 90^\circ$

Theorem 10-4: Inscribed Quadrilateral

A quadrilateral can be **inscribed in a circle** if and only if its **opposite angles are supplementary**.

$$m\angle A + m\angle C = 180^\circ \quad \text{and} \quad m\angle B + m\angle D = 180^\circ$$

Theorem 10-5: Tangent-Chord Angle

The measure of an angle formed by a **tangent and a chord** equals half the intercepted arc.

$$m\angle = \frac{1}{2}m(\widehat{\text{intercepted arc}})$$

Theorem 10-6: Two Chords — Interior Angle

Two **chords** intersecting **inside** a circle: the angle equals half the sum of the intercepted arcs.

$$m\angle 1 = \frac{1}{2}(m\widehat{AC} + m\widehat{BD})$$

Theorem 10-7: Two Secants/Tangents — Exterior Angle

Two **secants**, a **secant and a tangent**, or two **tangents** from an external point: the angle equals half the positive difference of the intercepted arcs.

$$m\angle P = \frac{1}{2}(m\widehat{\text{far}} - m\widehat{\text{near}})$$

Angle Formed By Summary

Position	Formula
Center (central $\angle$)	$m\angle = m\widehat{\text{arc}}$
On circle (inscribed)	$m\angle = \frac{1}{2}m\widehat{\text{arc}}$
Inside circle (2 chords)	$m\angle = \frac{1}{2}(\widehat{\text{arc}}_1 + \widehat{\text{arc}}_2)$
Outside circle	$m\angle = \frac{1}{2}(\widehat{\text{far}} - \widehat{\text{near}})$

Theorem 10-8: Two Chords — Segments

When two **chords** intersect inside a circle:

$$EA \cdot EB = EC \cdot ED$$

(Products of chord segments are equal.)

Theorem 10-9: Two Secants from External Point

Two **secants** from an external point P :

$$PA \cdot PB = PC \cdot PD$$

(whole secant $\times$ external segment)

Theorem 10-10: Secant and Tangent from External Point

A **secant** and a **tangent** from external point P (tangent length = t):

$$t^2 = PA \cdot PB$$

(tangent² = whole secant × external segment)

Example — Inscribed Angle

If $m\widehat{AC} = 80^\circ$, find the inscribed angle $\angle ABC$.

$$m\angle ABC = \frac{1}{2}(80^\circ) = 40^\circ$$

Chapter 10 – Theorem Summary

#	Name	Key Idea
Thm 10-1	Tangent $\perp$ Radius	Tangent $\perp$ radius at tangency
Thm 10-2	Two Tangents	External tangent segs. $\cong$
Thm 10-3	Inscribed Angle	$m\angle = \frac{1}{2}\widehat{\text{arc}}$
Cor 10-1	Same Arc	Same arc $\rightarrow$ inscribed $\angle$ s $\cong$
Cor 10-2	Semicircle	Inscribed in semicircle = 90°
Thm 10-4	Inscribed Quad	Opp. $\angle$ s supp.
Thm 10-6	Chords Interior	$\frac{1}{2}(\text{arc}_1 + \text{arc}_2)$
Thm 10-7	External Angle	$\frac{1}{2}(\text{far} - \text{near})$
Thm 10-8	Chord Segments	$EA \cdot EB = EC \cdot ED$
Thm 10-9	Secant Segments	$PA \cdot PB = PC \cdot PD$
Thm 10-10	Tangent-Secant	$t^2 = \text{whole} \times \text{external}$